

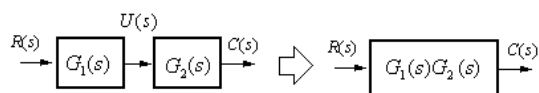
结构图等效变换

1. 环节串联:

$$C = G_2(s) \cdot U$$

$$\downarrow U = G_1(s) \cdot R$$

$$= G_2(s) \cdot G_1(s) \cdot R$$



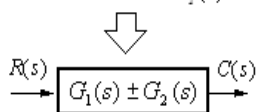
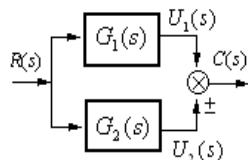
2. 环节并联

$$C = U_1 \pm U_2$$

$$\downarrow U_1 = G_1(s) \cdot R$$

$$\downarrow U_2 = G_2(s) \cdot R$$

$$= G_1(s)R \pm G_2(s)R = [G_1(s) \pm G_2(s)] \cdot R$$



3. 反馈连接

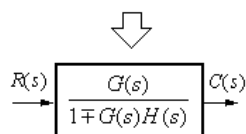
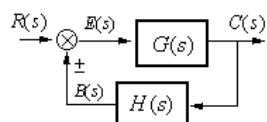
$$C = G(s) \cdot E$$

$$\downarrow E = R \pm B = R \pm H(s) \cdot C$$

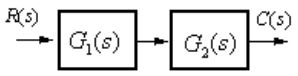
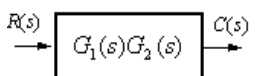
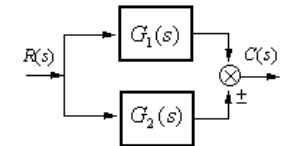
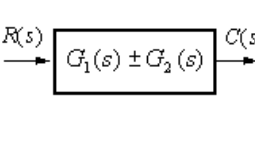
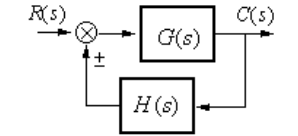
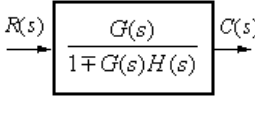
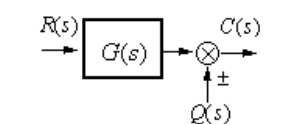
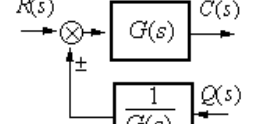
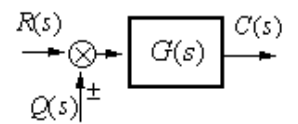
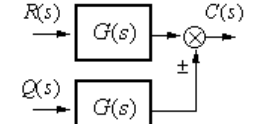
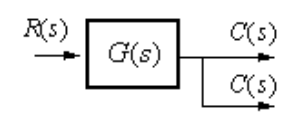
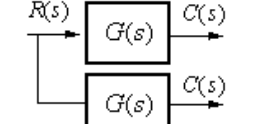
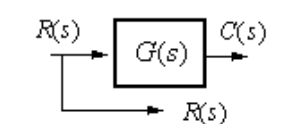
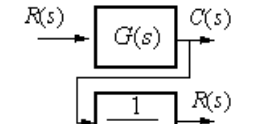
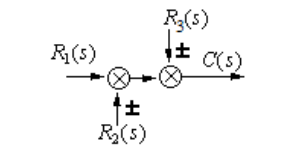
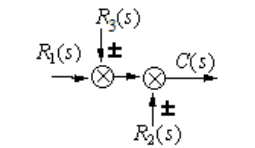
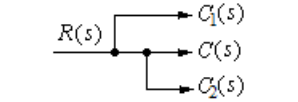
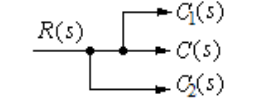
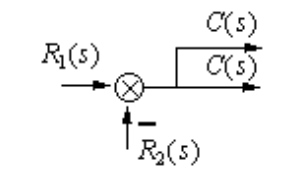
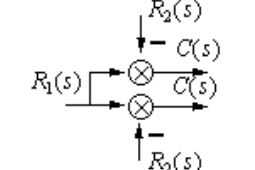
$$C = G(s) \cdot R \pm G(s)H(s) \cdot C$$

$$[1 \mp G(s)H(s)]C = G(s) \cdot R$$

$$\Phi(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 \mp G(s)H(s)}$$



结构图等效变换规则

变换方式	原方框图	等效方框图	等效运算关系
串联			$C(s) = G_1(s)G_2(s)R(s)$
并联			$C(s) = [G_1(s) \pm G_2(s)]R(s)$
反馈			$C(s) = \frac{G(s)R(s)}{1 + G(s)H(s)}$
比较点前移			$C(s) = R(s)G(s) \pm Q(s)$ $= [R(s) \pm \frac{Q(s)}{G(s)}]G(s)$
比较点后移			$C(s) = [R(s) \pm Q(s)]G(s)$ $= R(s)G(s) \pm Q(s)G(s)$
引出点前移			$C(s) = G(s)R(s)$
引出点后移			$R(s) = R(s)G(s) \frac{1}{G(s)}$ $C(s) = G(s)R(s)$
比较点之间的移动			$C(s) = R_1(s) \pm R_2(s) \pm R_3(s)$
引出点之间的移动			$C(s) = C_1(s) = C_2(s) = R(s)$
比较点与引出点之间的移动			$C(s) = R_1(s) - R_2(s)$